The reduced stated SL_n-skein algebra of the punctured bigon and the generalized quantum double

Abstract

We construct an explicit algebra isomorphism between the generalized quantum double $\mathcal{D} = U_q(\mathfrak{b}_-) \bowtie U_q(\mathfrak{b}_+)$ of the Borel subalgebras of the simply-connected $U_q(\mathfrak{sl}_n)$, for the Drinfeld–Tanisaki skew pairing, and the reduced stated SL_n-skein algebra $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$ of a bigon with one interior puncture: the two families of horizontal arcs of $\mathfrak{B}^*$ become a lower and an upper triangular quantum matrix, whose diagonal entries generate the two copies of the weight torus and whose off-diagonal entries are root vectors up to toral factors. We deduce that $U_q(\mathfrak{sl}_n)$ is the quotient of $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$ by the relations $b_{aa}c_{aa} = 1$ between the diagonal entries. For $n = 2$ both statements are due to Korinman. The proof combines the Lê–Sikora factorization of stated skein algebras with the Faddeev–Reshetikhin–Takhtajan realization of quantum Borel algebras by triangular quantum matrices, with all normalizations made explicit.

1 Introduction

1.1 The problem

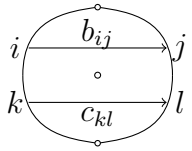
Stated skein algebras attach to a surface with marked boundary an algebra spanned by framed oriented tangles ("webs") whose endpoints carry labels called states, modulo local skein relations; they were introduced for SL_2 by Lê [9], developed by Costantino and Lê [4], and defined for SL_n by Lê and Sikora [10]. For the *bigon* $\mathfrak{B}$, a closed disk with two ideal points removed from its boundary, the stated SL_n-skein algebra is the quantum coordinate algebra $\mathcal{O}_q(SL_n)$ [10, Theorem 6.3]. The problem treated here concerns the *punctured bigon* $\mathfrak{B}^*$, obtained from $\mathfrak{B}$ by removing one interior point. It asks: (1) to construct an explicit algebra isomorphism between $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$, the reduced stated SL_n-skein algebra of $\mathfrak{B}^*$ (Section 1.2), and the generalized quantum double $\mathcal{D} = U_q(\mathfrak{b}_-) \bowtie U_q(\mathfrak{b}_+)$ of the Borel subalgebras of the simply-connected quantum group $U_q(\mathfrak{sl}_n)$ (the algebra written $U_q^{ext}(\mathfrak{sl}(n))$ in [7, Section 8.5.3]), the double being taken with respect to the Drinfeld–Tanisaki skew pairing

$$\langle \hat{K}_i, \hat{K}_j \rangle = q^{\frac{1}{n} - \delta_{ij}}, \qquad \langle E_i, F_j \rangle = \frac{\delta_{ij}}{q^{-1} - q}, \qquad \langle \hat{K}_i, F_j \rangle = \langle E_j, \hat{K}_i \rangle = 0, \tag{1}$$

where $\hat{K}_1, \dots, \hat{K}_n$ are the toral generators of the simply-connected Borel algebras (Section 2); (2) to deduce that $U_q(\mathfrak{sl}_n) \cong \overline{\mathcal{S}}_n(\mathfrak{B}^*)/I$, where I is the two-sided ideal generated by $b_{ii}c_{ii} - 1$, $1 \le i \le n$, and b_{ij}, c_{kl} are the two horizontal rightward arcs of $\mathfrak{B}^*$ with the indicated states (Section 1.2).

1.2 Interpretation of the statement

The statement we received does not display the diagram of the arcs b_{ij}, c_{kl} (only an undefined macro stands in its place) or define its skein notation. We interpret the algebra in the question as the *reduced* stated SL_n-skein algebra $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$ of Lê–Yu [11, Section 7.1], the quotient of the stated skein algebra $\mathcal{S}_n(\mathfrak{B}^*)$ by the two-sided ideal generated by the *bad corner arcs* recalled in Section 3.1; the unreduced algebra contains two full copies of $\mathcal{O}_q(SL_n)$ (Section 3) and is much larger than the double. We fix the missing diagram as follows. The surface $\mathfrak{B}^*$ has its usual planar orientation, its two ideal points at the top and bottom, its two boundary edges on the left and right, and the puncture in the middle; b_{ij} passes above the puncture and c_{kl} below it. Both arcs point to the right, and the first index is the state at the left endpoint:



The arc b_{ij} cuts off a disk containing only the upper ideal point, i.e. it is a *corner arc* there, and c_{kl} is a corner arc at the lower ideal point. At the upper corner a rightward arc is bad when its left state is smaller than its right state; at the lower corner the inequality is reversed (the general rule is recalled in (S4), Section 3.1). Consequently, in the reduced algebra,

$$b_{ij} = 0 \ (i < j), \qquad c_{kl} = 0 \ (k > l),$$

so the matrix $B = (b_{ij})$ is lower triangular and $C = (c_{kl})$ is upper triangular. If the intended diagram names the two arcs in the opposite order, the two families must be renamed accordingly in all formulas below.

1.3 The main result and the structure of the proof

Write H_a^- and H_a^+ $(1 \le a \le n)$ for the two copies of the toral generators $\hat{K}_a$ in $U_q(\mathfrak{b}_-)$ and $U_q(\mathfrak{b}_+)$, E_i, F_i $(1 \le i < n)$ for the Chevalley generators, and $s = q - q^{-1}$. Informally, the answer to (1) is that the assignment $b_{aa} \mapsto (H_a^-)^{-1}$, $c_{aa} \mapsto H_a^+$, $b_{i+1,i} \mapsto -s\, F_i (H_{i+1}^-)^{-1}$, $c_{i,i+1} \mapsto s\, H_{i+1}^+ E_i$ extends to an algebra isomorphism $\Phi : \overline{\mathcal{S}}_n(\mathfrak{B}^*) \to U_q(\mathfrak{b}_-) \bowtie U_q(\mathfrak{b}_+)$, under which the remaining arcs b_{ji} and c_{ij} with $j - i \ge 2$ go to root vectors for the roots $\pm(\epsilon_i - \epsilon_j)$ of $\mathfrak{sl}_n$ times toral factors, and the ideal I goes to the ideal identifying the two tori, so that $\overline{\mathcal{S}}_n(\mathfrak{B}^*)/I \cong U_q(\mathfrak{sl}_n)$. The precise statement, with all conventions, is Theorem 2.2. The proof has four inputs, assembled in Section 5. (i) *A presentation of the double* (Proposition 2.1): $\mathcal{D}$ is generated by the two Borel algebras subject to explicit mixed relations, so a homomorphism

out of $\mathcal{D}$ only has to be checked on Chevalley generators. (ii) *A presentation of the skein algebra* (Lemma 3.1): $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$ is generated by the entries of B and C subject to the relations $RB_1B_2 = B_2B_1R$, $RC_1C_2 = C_2C_1R$, $RB_1C_2 = C_2B_1R$ of RTT type (R the standard R-matrix, Section 3.2) and the triangularity and determinant conditions, and multiplication is a linear isomorphism $A_\ell \otimes A_u \cong \overline{\mathcal{S}}_n(\mathfrak{B}^*)$, where A_ℓ, A_u are the triangular quantum matrix algebras (Section 3.2). It is deduced from the Lê–Sikora factorization [10] of the *unreduced* algebra by an explicit computation showing that the bad-arc ideal is compatible with that factorization. (iii) *Triangular quantum matrices are quantum Borel algebras* (Lemma 4.1), the Faddeev–Reshetikhin–Takhtajan realization [6] with precise normalizations; injectivity is proved by comparing Poincaré–Birkhoff–Witt (PBW) bases, i.e. bases of ordered monomials, on the two sides. (iv) *Verification of the mixed relations* in $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$, a short computation with the entries of R (Section 5, Steps 2–3). We in fact construct $\Psi = \Phi^{-1}$ first, since (i) reduces its existence to finitely many relations; bijectivity then follows from the three established linear isomorphisms, because Ψ is, on underlying vector spaces, the composite of the triangular decompositions $\mathcal{D} \cong U_q(\mathfrak{b}_-) \otimes U_q(\mathfrak{b}_+) \cong A_\ell \otimes A_u \cong \overline{\mathcal{S}}_n(\mathfrak{B}^*)$. The formulas for Φ on all arcs follow by a recursion inside the triangular matrices, and part (2) is a direct consequence (Section 6).

1.4 Relation to the literature

For $n = 2$ the content of Theorem 2.2 is known: Korinman [8, Theorems 1.2 and 2.20] identifies the reduced stated SL_2-skein algebra of the punctured bigon with the Drinfeld double of the quantum Borel algebra, as a Hopf algebra, and obtains the simply-connected quantum $\mathfrak{sl}_2$ as the quotient by $K^{1/2}L^{1/2} - 1$ (in Korinman's notation for the two toral generators), our relation $b_{aa}c_{aa} = 1$. A diagrammatic precursor is Bigelow's definition of $U_q(\mathfrak{sl}_2)$ [1]; the unreduced punctured bigon and its exchange relations are treated for SL_2 in [4, Section 4.8], where the factorization-and-exchange architecture used below originates. We are not aware of a published statement of Theorem 2.2 for general n (Lê and Yu [11, Section 1.9] announce related work of Lê and Sikora on the punctured bigon); it is an explicit higher-rank version of an established rank-one identification, assembled from standard ingredients cited where they are used. The reduction argument in Lemma 3.1 is the punctured-bigon case of the mechanism of [11, Theorem 7.2], made explicit.

1.5 Conventions

Throughout, $n \geq 2$ and $\mathbb{K} = \mathbb{Q}(q^{1/(2n)})$ with q an indeterminate; all algebras are unital associative $\mathbb{K}$-algebras, $\otimes = \otimes_{\mathbb{K}}$, and $[x, y] = xy - yx$. We abbreviate $s = q - q^{-1}$, which is invertible, as the pairing (1) requires. Quantum groups are the unrestricted, simply-connected ones over $\mathbb{K}$. We write $\mathbf{1}_{i<j}$, $\mathbf{1}_{i>k}$, etc. for the truth value (0 or 1) of the indicated condition on indices, and δ_{ij} for the Kronecker delta. Indices a, b, and all matrix and state indices, run over $\{1, \dots, n\}$; the index i of a Chevalley generator E_i, F_i (and of the K_i of Section 2) runs over $\{1, \dots, n-1\}$; root vectors and off-diagonal matrix entries are indexed by pairs $1 \leq i < j \leq n$.

2 Borel algebras, the skew pairing and the double

This section fixes the quantum-group side. We recall the simply-connected Borel algebras, the Hopf structures and skew pairing that enter the problem, and the generalized quantum double; we then derive a presentation of the double (Proposition 2.1) and state the main theorem.

2.1 The simply-connected Borel algebras

Put

$$\eta_{ai} = \delta_{ai} - \delta_{a,i+1} \qquad (1 \le a \le n,\ 1 \le i < n),$$

so that $(\eta_{1i}, \dots, \eta_{ni})$ is the coordinate vector of the simple root $\alpha_i = \epsilon_i - \epsilon_{i+1}$ of $\mathfrak{sl}_n$, where $\epsilon_1, \dots, \epsilon_n$ are the coordinate functionals on the diagonal matrices. The *positive Borel algebra* $U_q(\mathfrak{b}_+)$ is the algebra generated by $(H_a^+)^{\pm 1}$ $(1 \le a \le n)$ and E_i $(1 \le i < n)$ subject to

$$H_a^+ H_b^+ = H_b^+ H_a^+, \qquad \textstyle\prod_{a=1}^n H_a^+ = 1, \tag{2}$$

$$H_a^+ E_i (H_a^+)^{-1} = q^{\eta_{ai}} E_i, \tag{3}$$

$$E_i E_j = E_j E_i \ \ (|i-j| > 1), \qquad E_i^2 E_j - (q + q^{-1}) E_i E_j E_i + E_j E_i^2 = 0 \ \ (|i-j| = 1). \tag{4}$$

The *negative Borel algebra* $U_q(\mathfrak{b}_-)$ is generated by $(H_a^-)^{\pm 1}$ and F_i subject to (2) for the H_a^-, to

$$H_a^- F_i (H_a^-)^{-1} = q^{-\eta_{ai}} F_i, \tag{5}$$

and to the Serre relations (4) with E replaced by F. We write $K_i^{\pm} = H_i^{\pm}(H_{i+1}^{\pm})^{-1}$, so that $K_i^+ E_j (K_i^+)^{-1} = q^{a_{ij}} E_j$ and $K_i^- F_j (K_i^-)^{-1} = q^{-a_{ij}} F_j$ with (a_{ij}) the Cartan matrix of $\mathfrak{sl}_n$. The subalgebra generated by the $H_a^{\pm}$ is the group algebra of $\mathbb{Z}^n/\mathbb{Z}(1, \dots, 1)$, the weight lattice of $\mathfrak{sl}_n$ ($H_a^{\pm}$ corresponding to the class of ϵ_a), whereas the adjoint version would contain only the root lattice spanned by the $K_i^{\pm}$; this is what "simply connected" refers to.

The simply-connected quantum group $U_q(\mathfrak{sl}_n)$ is generated by $H_a^{\pm 1}$, E_i, F_i subject to (2), to $H_a E_i H_a^{-1} = q^{\eta_{ai}} E_i$, $H_a F_i H_a^{-1} = q^{-\eta_{ai}} F_i$, to the Serre relations for the E_i and for the F_i, and to

$$[E_i, F_j] = \delta_{ij} \frac{K_i - K_i^{-1}}{s}, \qquad K_i = H_i H_{i+1}^{-1}. \tag{6}$$

2.2 Hopf structure, skew pairing and the generalized double

Both Borel algebras are Hopf algebras, with the $H_a^{\pm}$ group-like, counit $\varepsilon(E_i) = \varepsilon(F_i) = 0$, and (as in [7, Chapter 6])

$$\Delta(E_i) = E_i \otimes K_i^+ + 1 \otimes E_i, \qquad \Delta(F_i) = F_i \otimes 1 + (K_i^-)^{-1} \otimes F_i. \tag{7}$$

We use Sweedler's notation $\Delta(a) = \sum a_{(1)} \otimes a_{(2)}$. A *skew pairing* of $U_q(\mathfrak{b}_+)$ with $U_q(\mathfrak{b}_-)$ is a bilinear form $\langle\ ,\ \rangle : U_q(\mathfrak{b}_+) \times U_q(\mathfrak{b}_-) \to \mathbb{K}$ such that $\langle 1, b\rangle = \varepsilon(b)$, $\langle a, 1\rangle = \varepsilon(a)$ and

$$\langle aa', b\rangle = \sum \langle a, b_{(1)}\rangle \langle a', b_{(2)}\rangle, \qquad \langle a, bb'\rangle = \sum \langle a_{(1)}, b'\rangle \langle a_{(2)}, b\rangle. \tag{8}$$

By (8), a skew pairing is determined by its values on pairs of generators. The pairing in the problem is the *Drinfeld–Tanisaki pairing* ([7, Section 6.3.1], [13]; see also [5, Section 13]), whose values on generators are, in our notation,

$$\langle H_a^+, H_b^- \rangle = q^{1/n - \delta_{ab}}, \qquad \langle E_i, F_j \rangle = -\frac{\delta_{ij}}{s}, \qquad \langle H_a^+, F_i \rangle = \langle E_i, H_a^- \rangle = 0. \tag{9}$$

The exponent $\frac{1}{n} - \delta_{ab}$ is $-(\bar\epsilon_a, \bar\epsilon_b)$ for the invariant form on the weight lattice of $\mathfrak{sl}_n$, $\bar\epsilon_a$ being the class of ϵ_a. Two consequences of (8) for group-like elements H, H' will be used constantly: the pairing is multiplicative in each group-like variable, and $\langle H, (H')^{-1} \rangle = \langle H, H' \rangle^{-1}$.

The *generalized quantum double* (Drinfeld [5, Section 13]; [7, Section 8.2.1]) of this skew-paired pair is the algebra

$$\mathcal{D} = U_q(\mathfrak{b}_-) \bowtie U_q(\mathfrak{b}_+)$$

whose underlying vector space is $U_q(\mathfrak{b}_-) \otimes U_q(\mathfrak{b}_+)$, which contains $U_q(\mathfrak{b}_-) = U_q(\mathfrak{b}_-) \otimes 1$ and $U_q(\mathfrak{b}_+) = 1 \otimes U_q(\mathfrak{b}_+)$ as subalgebras, in which $w \cdot u = w \otimes u$ for $w \in U_q(\mathfrak{b}_-)$, $u \in U_q(\mathfrak{b}_+)$, and in which the products in the other order are governed by the *crossing rule*

$$\sum \langle u_{(1)}, w_{(1)} \rangle \, u_{(2)} w_{(2)} = \sum \langle u_{(2)}, w_{(2)} \rangle \, w_{(1)} u_{(1)} \qquad (u \in U_q(\mathfrak{b}_+),\ w \in U_q(\mathfrak{b}_-)). \tag{10}$$

That these requirements define an associative algebra, in which the multiplication map $U_q(\mathfrak{b}_-) \otimes U_q(\mathfrak{b}_+) \to \mathcal{D}$, $w \otimes u \mapsto wu$, is a linear isomorphism, is the basic theorem on the double [7, Section 8.2.1]; only its algebra structure is relevant here. The form (10) of the crossing rule is the one of [7, Section 8.2.1] for a skew pairing satisfying (8); as a consistency check, identifying the two tori in $\mathcal{D}$ returns $U_q(\mathfrak{sl}_n)$ with its standard relation (6) (Section 6).

2.3 A presentation of the double

Proposition 2.1. *The algebra $\mathcal{D}$ is generated by $(H_a^\pm)^{\pm 1}$, E_i, F_i subject to the defining relations of $U_q(\mathfrak{b}_+)$ and of $U_q(\mathfrak{b}_-)$ together with the mixed relations*

$$H_a^- H_b^+ = H_b^+ H_a^-, \tag{11}$$

$$H_a^- E_i = q^{\eta_{ai}} E_i H_a^-, \qquad H_a^+ F_i = q^{-\eta_{ai}} F_i H_a^+, \tag{12}$$

$$[E_i, F_j] = \delta_{ij} \, \frac{K_i^+ - (K_i^-)^{-1}}{s}. \tag{13}$$

Proof. The relations hold in $\mathcal{D}$. Apply (10) to $u = E_i$, $w = F_j$. By (7), the terms of the left side are indexed by the four pairs of tensor factors; the pairings $\langle E_i, (K_j^-)^{-1} \rangle$ and $\langle 1, F_j \rangle = \varepsilon(F_j)$ vanish, while $\langle 1, (K_j^-)^{-1} \rangle = 1$, so the left side is $\langle E_i, F_j \rangle K_i^+ + E_i F_j$. On the right side only $\langle K_i^+, 1 \rangle F_j E_i$ and $\langle E_i, F_j \rangle (K_j^-)^{-1}$ survive. Thus $-\frac{\delta_{ij}}{s} K_i^+ + E_i F_j = F_j E_i - \frac{\delta_{ij}}{s} (K_i^-)^{-1}$, which is (13). For $u = H_a^+$, $w = F_i$ the rule reads $\langle H_a^+, (K_i^-)^{-1} \rangle H_a^+ F_i = F_i H_a^+$, and multiplicativity gives $\langle H_a^+, (K_i^-)^{-1} \rangle = \langle H_a^+, H_i^- \rangle^{-1} \langle H_a^+, H_{i+1}^- \rangle = q^{\delta_{ai} - \delta_{a,i+1}} = q^{\eta_{ai}}$, which is the second relation in (12). For $u = E_i$, $w = H_a^-$ the rule reads $E_i H_a^- = \langle K_i^+, H_a^- \rangle H_a^- E_i$ with $\langle K_i^+, H_a^- \rangle = q^{-\eta_{ai}}$, which is the first relation. For $u = H_b^+$, $w = H_a^-$ both sides carry the same nonzero scalar $\langle H_b^+, H_a^- \rangle$, which cancels to give (11).

The relations are complete. Let P be the algebra presented by the two Borel presentations and (11)–(13), and let $\pi : P \to \mathcal{D}$ be the canonical map; it is surjective because $\mathcal{D}$ is generated by the two Borel subalgebras. We claim that P is spanned by the products wu with $w \in U_q(\mathfrak{b}_-)$, $u \in U_q(\mathfrak{b}_+)$. In a word in the generators, call a pair consisting of a positive letter ($(H_a^+)^{\pm 1}$, E_i) and a negative letter ($(H_a^-)^{\pm 1}$, F_j) standing anywhere to its right an inversion. If a word has an inversion, it has one with the two letters adjacent, and the mixed relations swap them: a toral swap only introduces a scalar and lowers the number of inversions by one, while (13) replaces E_iF_j by F_jE_i (one inversion fewer) plus a toral word with fewer Chevalley letters. By induction on the number of Chevalley letters, and for a fixed number of them on the number of inversions, every word becomes a linear combination of products wu. Hence $\mu : U_q(\mathfrak{b}_-) \otimes U_q(\mathfrak{b}_+) \to P$, $w \otimes u \mapsto wu$, is surjective, while $\pi \circ \mu$ is the multiplication map of $\mathcal{D}$, a linear isomorphism. So μ is bijective, π is injective, and $P \cong \mathcal{D}$. □

2.4 Root vectors and the main theorem

For $1 \le i < j \le n$ define root vectors recursively by

$$E_{i,i+1} = E_i, \qquad E_{ij} = E_{i,j-1}E_{j-1} - q^{-1}E_{j-1}E_{i,j-1} \qquad (j > i+1), \tag{14}$$

$$F_{i+1,i} = F_i, \qquad F_{ji} = F_{j-1}F_{j-1,i} - qF_{j-1,i}F_{j-1} \qquad (j > i+1). \tag{15}$$

Thus E_{ij} has weight $\epsilon_i - \epsilon_j$ under both tori, i.e. $H_a^{\pm} E_{ij} (H_a^{\pm})^{-1} = q^{\delta_{ai} - \delta_{aj}} E_{ij}$ by (3) and (12), and F_{ji} has weight $\epsilon_j - \epsilon_i$. These normalizations make every off-diagonal entry of B and C a single root vector times a toral factor, with the uniform scalar $-s$, respectively s, independent of $j - i$ (see (17)).

Theorem 2.2. *Let $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$ be the reduced stated SL_n-skein algebra of the punctured bigon, with the arcs b_{ij}, c_{kl} of Section 1.2, and let $\mathcal{D} = U_q(\mathfrak{b}_-) \bowtie U_q(\mathfrak{b}_+)$ be the generalized quantum double of Section 2. An algebra isomorphism $\Phi : \overline{\mathcal{S}}_n(\mathfrak{B}^*) \to \mathcal{D}$ is given by*

$$\Phi(b_{aa}) = (H_a^-)^{-1}, \qquad \Phi(c_{aa}) = H_a^+, \tag{16}$$

$$\Phi(b_{ji}) = -s\, F_{ji} (H_j^-)^{-1}, \qquad \Phi(c_{ij}) = s\, H_j^+ E_{ij} \qquad (i < j), \tag{17}$$

the remaining entries being zero in $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$: $b_{ij} = c_{ji} = 0$ for $i < j$. Its inverse Ψ is given on the generators of $\mathcal{D}$ by

$$\Psi(H_a^-) = b_{aa}^{-1}, \qquad \Psi(H_a^+) = c_{aa}, \tag{18}$$

$$\Psi(F_i) = -s^{-1} b_{i+1,i}\, b_{i+1,i+1}^{-1}, \qquad \Psi(E_i) = s^{-1} c_{i+1,i+1}^{-1}\, c_{i,i+1}. \tag{19}$$

If $I \subset \overline{\mathcal{S}}_n(\mathfrak{B}^*)$ is the two-sided ideal generated by $b_{aa}c_{aa} - 1$, $1 \le a \le n$, then Φ induces an isomorphism

$$\overline{\mathcal{S}}_n(\mathfrak{B}^*)/I \;\cong\; U_q(\mathfrak{sl}_n)$$

with the simply-connected quantum group, under which H_a (the $\hat{K}_a$ of the problem), E_i, F_i correspond to the classes of c_{aa} ($= b_{aa}^{-1}$ modulo I) and of the right-hand sides of (19).

The diagonal entries b_{aa}, c_{aa} are invertible in $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$ by Lemma 3.1 below, so the formulas make sense. The proof occupies Sections 3–6, in the order (ii), (iii), (iv), part (2) of Section 1.3.

3 The reduced punctured bigon

3.1 Stated skein algebras

We recall what is needed from [10, 11]. A punctured bordered surface is $\mathfrak{S} = \overline{\mathfrak{S}} \setminus \mathcal{P}$, where $\overline{\mathfrak{S}}$ is a compact oriented surface and $\mathcal{P}$ is a finite set meeting every boundary component; the points of $\mathcal{P}$ on the boundary are the *ideal points*, those in the interior the *punctures*, and the components of $\partial\mathfrak{S}$ are the *boundary edges*. Thus $\mathfrak{B}^*$ has two ideal points, two boundary edges and one puncture. The stated SL_n-skein algebra $\mathcal{S}_n(\mathfrak{S})$ of [10] is the $\mathbb{K}$-module spanned by isotopy classes of *stated n-webs* in $\mathfrak{S} \times (-1,1)$, framed oriented tangles (more generally webs) with endpoints on $\partial\mathfrak{S} \times (-1,1)$ carrying *states* in $\{1,\dots,n\}$, modulo the SL_n-skein relations of [10]. The product xy places x above y, i.e. at greater height in the $(-1,1)$ factor. Relative heights of endpoints on the same boundary edge matter; for a single arc joining two different boundary edges no such choice arises, so b_{ij} and c_{kl} are determined by their isotopy class in $\mathfrak{B}^*$, their orientation and their states. (The algebras of [10] are defined over any commutative ring with an invertible $2n$-th root of q, and all coefficients below lie in $\mathbb{K}$; we work over $\mathbb{K}$ throughout.)

We use the following four facts.

(S1) (*Bigon* [10, Theorem 6.3]; conventions as in [11, Sections 3.1 and 4.7].) $\mathcal{S}_n(\mathfrak{B}) \cong \mathcal{O}_q(SL_n)$ as algebras, the rightward arc with left state i and right state j corresponding to the matrix generator t_{ij}; the matrix $T = (t_{ij})$ is a q-matrix in the sense of Section 3.2 below, with quantum determinant 1. The reversed arc is related to the antipode by [11, equation (83)]:

$$S(t_{ij}) = (-q)^{i-j} \overleftarrow{t}_{\bar{j},\bar{i}}, \qquad \bar{i} = n+1-i, \tag{20}$$

where $\overleftarrow{t}_{kl}$ denotes the leftward arc with left state k and right state l. Only the fact that the scalar is nonzero will matter.

(S2) (*Factorization along saturated arcs* [10, Theorem 8.8, Corollaries 8.14–8.15].) Let every component of $\mathfrak{S}$ have nonempty boundary, and let $\alpha_1,\dots,\alpha_m$ be a saturated system of disjoint proper arcs in $\mathfrak{S}$ (each complementary region contains exactly one ideal point or puncture, and the system is maximal with this property [10, Section 8.4]), each α_r joining two distinct boundary edges. Then a thickening of each α_r is a bigon whose algebra $A_{\alpha_r} \cong \mathcal{O}_q(SL_n)$ maps to $\mathcal{S}_n(\mathfrak{S})$ by an algebra homomorphism, and, for any fixed order of the factors, multiplication $A_{\alpha_1} \otimes \cdots \otimes A_{\alpha_m} \to \mathcal{S}_n(\mathfrak{S})$ is a linear isomorphism. In general this is only a linear isomorphism; no commutation between different factors is asserted.

(S3) (*Height exchange* [11, Section 4.10].) A stated web with two consecutive endpoints on the same boundary edge, both oriented the same way (as for two rightward arcs), equals a linear combination of the webs obtained by interchanging the heights of these two endpoints, with the states permuted, the coefficients being entries of the normalized matrix $q^{-1/n}R$,

where R is the matrix of (21) below (or entries of its inverse); they are computed from the crossing evaluations [11, equations (85)–(86)] and the counit identity [11, equation (90)].
(S4) (*Bad arcs and the reduced algebra* [11, Section 7.1].) A corner arc at an ideal point v is an arc cutting off from $\mathfrak{S}$ a disk containing v and no other ideal point or puncture. Certain corner arcs, determined by the orientation and an inequality between the two states, are *bad*; for $\mathfrak{B}^*$ the rule is that a corner arc at the upper ideal point, of either orientation, is bad when its state at the left endpoint is smaller than its state at the right endpoint, and a corner arc at the lower ideal point is bad when it is larger. The reduced algebra $\overline{\mathcal{S}}_n(\mathfrak{S})$ is the quotient of $\mathcal{S}_n(\mathfrak{S})$ by the two-sided ideal generated by all bad arcs.

3.2 Quantum matrices

Let $v_1, \dots, v_n$ be the standard basis of $\mathbb{K}^n$, let e_{ab} be the matrix units of $\mathrm{End}(\mathbb{K}^n)$, and put

$$R = \sum_{a,b=1}^{n} q^{\delta_{ab}} e_{aa} \otimes e_{bb} + s \sum_{a>b} e_{ab} \otimes e_{ba}, \qquad \text{so that} \qquad R(v_j \otimes v_l) = q^{\delta_{jl}} v_j \otimes v_l + s\, \mathbf{1}_{j<l}\, v_l \otimes v_j. \tag{21}$$

The R-matrix of the vector representation in the SL_n normalization of [11, Section 3.1] is $q^{-1/n}R$; the scalar $q^{-1/n}$ does not affect the relations $RT_1T_2 = T_2T_1R$ below and is cancelled explicitly where it occurs (Step 2 of Lemma 3.1). We write R^{ik}_{ab} for the entries of R, $R(v_a \otimes v_b) = \sum_{i,k} R^{ik}_{ab}\, v_i \otimes v_k$. For a matrix T with entries in an algebra put $T_1 = T \otimes \mathrm{id}$ and $T_2 = \mathrm{id} \otimes T$. A *q-matrix* is a matrix T with $RT_1T_2 = T_2T_1R$ [6]; in components (the entry in row (i,k) and column (j,l), computed with (21)),

$$q^{\delta_{ik}} t_{ij}t_{kl} + s\mathbf{1}_{i>k}\, t_{kj}t_{il} = q^{\delta_{jl}} t_{kl}t_{ij} + s\mathbf{1}_{j<l}\, t_{kj}t_{il}. \tag{22}$$

Equivalently, every 2×2 submatrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ satisfies

$$ab = qba, \quad ac = qca, \quad bd = qdb, \quad cd = qdc, \quad bc = cb, \quad ad - da = sbc. \tag{23}$$

The algebra $\mathcal{O}_q(SL_n)$ is generated by the entries of a universal q-matrix with the relation $\det_q T = 1$, where $\det_q T = \sum_{\sigma \in S_n} (-q)^{\ell(\sigma)} t_{1\sigma(1)} \cdots t_{n\sigma(n)}$, $\ell(\sigma)$ the number of inversions of σ [7, Section 9.2].

Let A_u (resp. A_ℓ) be the *universal upper* (resp. *lower*) *triangular q-matrix algebra*: it is generated by t_{ij} with $i \le j$ (resp. $i \ge j$), subject to the relations (23) of a q-matrix in which the other entries are set to 0, and to $\prod_a t_{aa} = 1$. We write $d_a = t_{aa}$. Three consequences of (23) will be used repeatedly. (Q1) The d_a commute: in the submatrix with rows and columns i, k one off-diagonal entry is forbidden, so $ad - da = sbc$ reads $d_i d_k = d_k d_i$; as their product is 1, each d_a is invertible. (Q2) In A_u the relations (23) give, for $i < j$, the first formula below, and in A_ℓ, for $i > j$, the second:

$$d_a t_{ij} = q^{\delta_{ai} - \delta_{aj}}\, t_{ij} d_a \quad (i < j), \qquad d_a t_{ij} = q^{\delta_{aj} - \delta_{ai}}\, t_{ij} d_a \quad (i > j). \tag{24}$$

(Q3) For $i < k < j$, the 2×2 submatrices with rows i, k and columns k, j in A_u, and with rows k, j and columns i, k in A_ℓ, give

$$[t_{ik}, t_{kj}] = s\, t_{ij} d_k \quad \text{in } A_u, \qquad [t_{ki}, t_{jk}] = s\, d_k t_{ji} \quad \text{in } A_\ell. \tag{25}$$

Since $\det_q$ of a triangular matrix is the product of its diagonal entries (only $\sigma = \mathrm{id}$ contributes), A_u and A_ℓ are the quotients of $\mathcal{O}_q(SL_n)$ by the strictly lower, respectively strictly upper, entries. Transposition $t_{ij} \mapsto t_{ji}$ is an algebra isomorphism $A_u \to A_\ell$, because (23) is invariant under $b \leftrightarrow c$.

3.3 Presentation of $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$

Lemma 3.1. *The algebra $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$ is generated by the entries of $B = (b_{ij})$ and $C = (c_{kl})$ subject to*

$$b_{ij} = 0 \ (i < j), \qquad c_{ij} = 0 \ (i > j), \qquad \prod_{a=1}^{n} b_{aa} = \prod_{a=1}^{n} c_{aa} = 1, \tag{26}$$

$$RB_1B_2 = B_2B_1R, \qquad RC_1C_2 = C_2C_1R, \qquad RB_1C_2 = C_2B_1R. \tag{27}$$

Moreover, the algebra homomorphisms $A_\ell \to \overline{\mathcal{S}}_n(\mathfrak{B}^*)$, $f \mapsto f(B)$, and $A_u \to \overline{\mathcal{S}}_n(\mathfrak{B}^*)$, $g \mapsto g(C)$, obtained by substituting $t_{ij} \mapsto b_{ij}$, respectively $t_{ij} \mapsto c_{ij}$ (well defined because of (26)–(27)), are injective, and the multiplication map

$$A_\ell \otimes A_u \longrightarrow \overline{\mathcal{S}}_n(\mathfrak{B}^*), \qquad f \otimes g \longmapsto f(B)g(C), \tag{28}$$

is a linear isomorphism. In particular the diagonal entries b_{aa}, c_{aa} are invertible in $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$.

The $(i,k;j,l)$ component of the mixed relation $RB_1C_2 = C_2B_1R$ is, by (22),

$$q^{\delta_{ik}} b_{ij}c_{kl} + s\mathbf{1}_{i>k}\, b_{kj}c_{il} = q^{\delta_{jl}} c_{kl}b_{ij} + s\mathbf{1}_{j<l}\, c_{kj}b_{il}. \tag{29}$$

Proof. Step 1: the unreduced factorization. A thin strip around either horizontal arc is a bigon, whose stated skein algebra is $\mathcal{O}_q(SL_n)$ by (S1); write A_B and A_C for the two copies, with generating matrices B and C. Each arc joins the two distinct boundary edges of $\mathfrak{B}^*$, so the inclusion of its strip induces algebra homomorphisms $A_B \to \mathcal{S}_n(\mathfrak{B}^*)$ and $A_C \to \mathcal{S}_n(\mathfrak{B}^*)$. The two arcs cut $\mathfrak{B}^*$ into two disks containing one boundary ideal point each and a disk containing the puncture, and any additional compact proper arc disjoint from these arcs separates one of the complementary regions and leaves a component containing neither a boundary ideal point nor an interior puncture. Thus the system is saturated, and (S2) gives a linear isomorphism

$$A_B \otimes A_C \xrightarrow{\sim} \mathcal{S}_n(\mathfrak{B}^*), \qquad f \otimes g \longmapsto f(B)g(C), \qquad A_B, A_C \cong \mathcal{O}_q(SL_n). \tag{30}$$

Step 2: the exchange relations. The first two relations in (27) hold in A_B and A_C by (S1). For the mixed relation, consider the two disjoint arcs with states (i,j) (upper arc) and (k,l) (lower arc) placed so that the c-end is higher at the left edge and the b-end is higher at the right edge; this configuration is neither of the products $b_{ij}c_{kl}$, $c_{kl}b_{ij}$. Applying (S3) at the left edge puts the b-end on top there and expresses the configuration as a combination of products $b_{pj}c_{rl}$ with coefficients $q^{-1/n}R^{ik}_{pr}$; applying (S3) at the right edge instead puts the c-end on

top there and expresses it as a combination of products $c_{kr}b_{ip}$ with coefficients $q^{-1/n}R^{pr}_{jl}$. By (21), apart from the common factor $q^{-1/n}$, the identity contribution has coefficient $q^{\delta_{ik}}$, respectively $q^{\delta_{jl}}$, and the only other contribution is the interchange of the two states, with coefficient s, which occurs when $i > k$, respectively $j < l$. Equating the two expressions and cancelling $q^{-1/n}$ gives exactly (29), i.e. $RB_1C_2 = C_2B_1R$. That the exchange coefficients are entries of $q^{-1/n}R$ rather than of its inverse is determined by the orientation and height conventions of [11, Section 4.10]; the quotient computation of Section 6 is consistent with this choice but does not establish it.

Step 3: the ideal of rightward bad arcs. Let $J_B \subset A_B$ be the two-sided ideal generated by the b_{ij} with $i < j$ and $J_C \subset A_C$ the one generated by the c_{kl} with $k > l$; their generators are the rightward bad arcs. Under (30) consider the subspace $N = J_B \otimes A_C + A_B \otimes J_C \subset \mathcal{S}_n(\mathfrak{B}^*)$. We claim that N is the two-sided ideal generated by the rightward bad arcs. Solving (29) for the out-of-order product, i.e. writing $C_2B_1 = RB_1C_2R^{-1}$ with $R^{-1} = \sum_{a,b} q^{-\delta_{ab}} e_{aa} \otimes e_{bb} - s\sum_{a>b} e_{ab} \otimes e_{ba}$, gives

$$\begin{aligned} c_{kl}b_{ij} = {} & q^{\delta_{ik}-\delta_{jl}}\, b_{ij}c_{kl} + s\mathbf{1}_{i>k}\, q^{-\delta_{jl}}\, b_{kj}c_{il} \\ & - s\mathbf{1}_{j<l}\, q^{\delta_{ik}}\, b_{il}c_{kj} - s^2\mathbf{1}_{i>k}\mathbf{1}_{j<l}\, b_{kl}c_{ij}. \end{aligned} \tag{31}$$

In every term on the right, the B-entry has row index $\le i$ and column index $\ge j$. Hence if b_{ij} is bad ($i < j$), so is the B-entry of every term; likewise the C-entry has row $\ge k$ and column $\le l$, so a bad C-entry stays bad. Hence if a word in the entries of B and C contains a bad entry and is reordered into the form (word in B)(word in C) by repeated use of (31), every summand of the result still contains a bad entry of the same family. Since J_B, J_C are ideals in their own factors, this gives $N \cdot A_B \subset N$ and $A_C \cdot N \subset N$, while $A_B \cdot N \subset N$ and $N \cdot A_C \subset N$ are clear. So N is a two-sided ideal containing the rightward bad arcs and contained in the ideal they generate, which proves the claim.

Step 4: the reversed bad arcs give nothing new. The bad arcs of the other orientation are the leftward corner arcs $\overleftarrow{b}_{kl}$ with $k < l$ at the upper corner and $\overleftarrow{c}_{kl}$ with $k > l$ at the lower one (in the notation of (S1), applied in the strips A_B and A_C). By (20) in A_B, $\overleftarrow{b}_{kl}$ is a nonzero scalar multiple of $S(b_{\bar{l},\bar{k}})$, the $(\bar{l},\bar{k})$ entry of the inverse matrix B^{-1}, and $k < l$ means $\bar{l} < \bar{k}$. In $A_B/J_B \cong A_\ell$ (Section 3.2) the matrix B is lower triangular with invertible diagonal entries (Q1), hence has a lower triangular inverse (write $B = D(1 + D^{-1}L)$ with D the diagonal part of B and $D^{-1}L$ strictly lower triangular, hence nilpotent). Thus the strictly upper entries of B^{-1} vanish modulo J_B, and every $\overleftarrow{b}_{kl}$ with $k < l$ lies in J_B. The same argument applies to $\overleftarrow{c}_{kl}$, $k > l$, since $(k,l) \mapsto (\bar{l},\bar{k})$ preserves both strict inequalities. Hence N is the full bad-arc ideal and $\overline{\mathcal{S}}_n(\mathfrak{B}^*) = \mathcal{S}_n(\mathfrak{B}^*)/N$.

Step 5: conclusion. By Section 3.2, $A_B/J_B \cong A_\ell$ and $A_C/J_C \cong A_u$. Passing to the quotient by N in (30), and using $(V \otimes W)/(V' \otimes W + V \otimes W') \cong V/V' \otimes W/W'$, gives the linear isomorphism (28); its restrictions to $A_\ell \otimes 1$ and $1 \otimes A_u$ are the algebra homomorphisms $f \mapsto f(B)$ and $g \mapsto g(C)$, which are therefore injective. Finally, let P be the algebra presented by (26)–(27). The relations hold in $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$, so there is a canonical surjection $P \to \overline{\mathcal{S}}_n(\mathfrak{B}^*)$; by (31) every word in P is a combination of products (word in B)(word in C),

so $A_\ell \otimes A_u \to P$ is surjective, and its composite with $P \to \overline{\mathcal{S}}_n(\mathfrak{B}^*)$ is the bijection (28). As in the proof of Proposition 2.1, $P \to \overline{\mathcal{S}}_n(\mathfrak{B}^*)$ is injective, and the presentation is complete. $\square$

4 Triangular quantum matrices and the Borel algebras

That the upper and lower triangular quantum matrices realize the two quantum Borel algebras is due to Faddeev, Reshetikhin and Takhtajan [6] (see also [7, Section 8.5]). Since Theorem 2.2 is about precise normalizations, including the order of the toral factors, we record the isomorphisms with proof; A_u, A_ℓ and $d_a = t_{aa}$ are as in Section 3.2.

Lemma 4.1. *There are algebra isomorphisms*

$$U_q(\mathfrak{b}_+) \xrightarrow{\sim} A_u, \qquad H_a^+ \longmapsto d_a, \qquad E_i \longmapsto X_i := s^{-1} d_{i+1}^{-1} t_{i,i+1}, \tag{32}$$

$$U_q(\mathfrak{b}_-) \xrightarrow{\sim} A_\ell, \qquad H_a^- \longmapsto d_a^{-1}, \qquad F_i \longmapsto -s^{-1} t_{i+1,i}\, d_{i+1}^{-1}. \tag{33}$$

Under (32) *the root vectors* (14) *map to*

$$E_{ij} \longmapsto s^{-1} d_j^{-1} t_{ij} \qquad (i<j). \tag{34}$$

Proof. The map (32) *is a homomorphism.* The relations (2) hold for the d_a by (Q1), and (3) follows from (24): d_a commutes with d_{i+1}^{-1} and $d_a t_{i,i+1} d_a^{-1} = q^{\delta_{ai}-\delta_{a,i+1}} t_{i,i+1}$. For $|i-j|>1$ the elements X_i, X_j commute: the diagonal factors commute with one another, each diagonal factor commutes with the off-diagonal entry occurring in the other X by (24), and $t_{i,i+1}$, $t_{j,j+1}$ lie in distinct rows and columns, so (23) gives $[t_{i,i+1}, t_{j,j+1}] = s\, t_{i,j+1} t_{j,i+1}$ (for $i<j$), where $t_{j,i+1}=0$ because $j > i+1$. For the adjacent Serre relations put $W_i = s^{-1} d_{i+2}^{-1} t_{i,i+2}$. Then

$$X_i X_{i+1} - q^{-1} X_{i+1} X_i = W_i, \qquad X_i W_i = q W_i X_i, \qquad W_i X_{i+1} = q X_{i+1} W_i. \tag{35}$$

For the first identity, move the toral factors to the left using (24) (d_{i+2} commutes with $t_{i,i+1}$, and $t_{i+1,i+2} d_{i+1}^{-1} = q\, d_{i+1}^{-1} t_{i+1,i+2}$), which turns the left side into $s^{-2} d_{i+1}^{-1} d_{i+2}^{-1} [t_{i,i+1}, t_{i+1,i+2}] = s^{-1} d_{i+1}^{-1} d_{i+2}^{-1}\, t_{i,i+2} d_{i+1} = s^{-1} d_{i+2}^{-1} t_{i,i+2} = W_i$ by (25) and because d_{i+1} commutes with $t_{i,i+2}$. The other two identities reduce in the same way to the row relation $t_{i,i+1}t_{i,i+2} = q t_{i,i+2} t_{i,i+1}$ and the column relation $t_{i,i+2}t_{i+1,i+2} = q t_{i+1,i+2} t_{i,i+2}$. Substituting the first identity of (35) into the second and third and expanding yields $X_i^2 X_{i+1} - (q+q^{-1}) X_i X_{i+1} X_i + X_{i+1} X_i^2 = 0$ and the same relation with i and $i+1$ interchanged, i.e. both adjacent Serre relations (4). Thus (32) defines a homomorphism $\phi : U_q(\mathfrak{b}_+) \to A_u$.

Surjectivity and the root vectors. By (25), $t_{ij} = s^{-1}[t_{i,j-1}, t_{j-1,j}] d_{j-1}^{-1}$ for $j > i+1$, so the diagonal and first superdiagonal entries generate A_u and ϕ is surjective. The computation just made for the first identity in (35), with $(i, i+1, i+2)$ replaced by $(i, j-1, j)$ and with the inductive hypothesis $\phi(E_{i,j-1}) = s^{-1} d_{j-1}^{-1} t_{i,j-1}$, gives (34).

Injectivity, via PBW bases. Order the pairs $i<j$ lexicographically. On the matrix side, the ordered monomials

$$\Big(\prod_{i<j}^{\text{lex}} t_{ij}^{m_{ij}}\Big) d_1^{r_1} \cdots d_{n-1}^{r_{n-1}}, \qquad m_{ij} \in \mathbb{Z}_{\geq 0}, \quad r_a \in \mathbb{Z}, \tag{36}$$

form a basis of A_u. This follows from [11, Proposition 3.10], which gives such a basis, for any linear order of the off-diagonal entries, for the lower triangular quotient, by transposition; there the diagonal exponents are classes modulo $(1,\dots,1)$, uniquely represented by $(r_1,\dots,r_{n-1},0)$, and since the d_a are invertible all integer exponents occur. On the quantum-group side, the monomials $\prod_{i<j}^{\mathrm{lex}} E_{ij}^{m_{ij}}$ form a basis of the subalgebra $U_q(\mathfrak{n}_+)$ generated by the E_i (PBW theorem in type A, Rosso [12], Yamane [14]). For the specific root vectors (14) we use the Gröbner–Shirshov basis of [3, Theorem 3.1] (going back to [2]), which by the Composition–Diamond lemma [3, Lemma 2.4] yields exactly the lexicographically ordered root monomials: in their $U_q(A_{n-1})$, writing the quantum parameter as v, put $v=q^{1/2}$; their positive Serre coefficient v^2+v^{-2} becomes $q+q^{-1}$, and their recursion $x_{ij}=v\,x_{i,j-1}E_{j-1}-v^{-1}E_{j-1}x_{i,j-1}$ gives $x_{ij}=v^{j-i-1}E_{ij}$ by induction, so the basis assertion is unchanged by this rescaling. Next, the torus acts on $U_q(\mathfrak{n}_+)$ by the algebra automorphisms (3): the Serre relations are homogeneous, and the action descends to $\mathbb{Z}^n/\mathbb{Z}(1,\dots,1)$ because $\sum_a \eta_{ai}=0$. Hence $U_q(\mathfrak{b}_+)$ is the smash product of $U_q(\mathfrak{n}_+)$ with the group algebra of $\mathbb{Z}^n/\mathbb{Z}(1,\dots,1)$, and

$$\Big(\prod_{i<j}^{\mathrm{lex}} E_{ij}^{m_{ij}}\Big)(H_1^+)^{r_1}\cdots(H_{n-1}^+)^{r_{n-1}},\qquad m_{ij}\in\mathbb{Z}_{\ge 0},\quad r_a\in\mathbb{Z}, \tag{37}$$

is a basis of $U_q(\mathfrak{b}_+)$.

Now apply ϕ to (37). By (34) the image is $\prod^{\mathrm{lex}}(s^{-1}d_j^{-1}t_{ij})^{m_{ij}}\cdot\prod_a d_a^{r_a}$. Moving all diagonal factors to the right using (24) only produces a nonzero scalar, and the exponent of d_a becomes r_a-M_a, where $M_a=\sum_{i<a}m_{ia}$ (so $M_1=0$); finally $d_n^{-M_n}=(d_1\cdots d_{n-1})^{M_n}$. So the image is a nonzero scalar times the basis monomial (36) with the same (m_{ij}) and with r_a replaced by $r_a-M_a+M_n$ for $a<n$. For fixed (m_{ij}) this is a bijection of $\mathbb{Z}^{n-1}$. Distinct basis elements therefore map to distinct basis elements, and ϕ is injective.

The lower triangular case. The assignment $H_a^-\mapsto(H_a^+)^{-1}$, $F_i\mapsto -q^{-1}E_i$ is an algebra isomorphism $U_q(\mathfrak{b}_-)\to U_q(\mathfrak{b}_+)$: it carries (5) to (3), and the Serre relations are homogeneous (any nonzero scalar in place of $-q^{-1}$ would do; this one makes the composite below come out as (33)). Composing it with (32) and with transposition $A_u\to A_\ell$ (Section 3.2) gives an isomorphism $U_q(\mathfrak{b}_-)\to A_\ell$ with $H_a^-\mapsto d_a^{-1}$ and $F_i\mapsto -q^{-1}s^{-1}d_{i+1}^{-1}t_{i+1,i}$, which equals $-s^{-1}t_{i+1,i}d_{i+1}^{-1}$ by (24) for A_ℓ: $d_{i+1}t_{i+1,i}=q^{-1}t_{i+1,i}d_{i+1}$. This is (33). □

5 Proof of Theorem 2.2: the isomorphism

We construct $\Psi:\mathcal{D}\to\overline{\mathcal{S}}_n(\mathfrak{B}^*)$ first, because Proposition 2.1 reduces its existence to finitely many relations, and then obtain Φ as its inverse. In $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$ abbreviate

$$d_a=b_{aa},\qquad h_a=c_{aa},\qquad x_i=b_{i+1,i},\qquad y_i=c_{i,i+1}$$

(so the d_a of Lemma 4.1 become d_a when the lemma is applied to B and h_a when it is applied to C).

Step 1: the two Borel algebras embed. By Lemma 3.1, C is an upper triangular q-matrix with $\prod_a h_a = 1$ and B a lower triangular one with $\prod_a d_a = 1$, and $g \mapsto g(C)$, $f \mapsto f(B)$ are injective algebra maps $A_u \to \overline{\mathcal{S}}_n(\mathfrak{B}^*)$, $A_\ell \to \overline{\mathcal{S}}_n(\mathfrak{B}^*)$. Composing with Lemma 4.1 gives injective algebra homomorphisms

$$\begin{aligned} U_q(\mathfrak{b}_+) \to \overline{\mathcal{S}}_n(\mathfrak{B}^*): \quad & H_a^+ \mapsto h_a, \quad E_i \mapsto e_i := s^{-1} h_{i+1}^{-1} y_i, \\ U_q(\mathfrak{b}_-) \to \overline{\mathcal{S}}_n(\mathfrak{B}^*): \quad & H_a^- \mapsto d_a^{-1}, \quad F_i \mapsto f_i := -s^{-1} x_i d_{i+1}^{-1}. \end{aligned}$$

These are the assignments (18)–(19). By Proposition 2.1, they define a homomorphism $\Psi : \mathcal{D} \to \overline{\mathcal{S}}_n(\mathfrak{B}^*)$ as soon as the mixed relations (11)–(13) hold for d_a^{-1}, h_a, e_i, f_i.

Step 2: the mixed relations between the arcs. Specializing (29) and using triangularity, we get

$$d_a h_b = h_b d_a, \tag{38}$$

$$d_a y_i = q^{-\eta_{ai}} y_i d_a, \qquad h_a x_i = q^{-\eta_{ai}} x_i h_a, \tag{39}$$

$$x_j y_i = q^{\delta_{j,i+1} - \delta_{j+1,i}}\, y_i x_j \qquad (i \neq j), \tag{40}$$

$$[x_i, y_i] = s\,(h_i d_{i+1} - d_i h_{i+1}). \tag{41}$$

Indeed, for $(i,k;j,l) = (a,r;a,r)$ the two s-terms of (29) contain b_{ra} with $r < a$ or c_{ra} with $r > a$, which vanish, leaving (38); in the same way $(a,r;a,r+1)$ gives $q^{\delta_{ar}} d_a y_r = q^{\delta_{a,r+1}} y_r d_a$, the first relation in (39), and $(r+1,a;r,a)$ gives the second. For $(i,k;j,l) = (t+1,r;t,r+1)$ the relation reads $q^{\delta_{t+1,r}} x_t y_r + s\mathbf{1}_{t+1>r}\, b_{rt} c_{t+1,r+1} = q^{\delta_{t,r+1}} y_r x_t + s\mathbf{1}_{t<r+1}\, c_{rt} b_{t+1,r+1}$, and by triangularity the s-terms vanish unless $r = t$. For $r \neq t$ this is (40) with $(i,j) = (r,t)$; for $r = t$ it reads $x_r y_r + s\, d_r h_{r+1} = y_r x_r + s\, h_r d_{r+1}$, which is (41).

Step 3: the mixed relations of the double. Relations (38)–(39), together with the fact that d_a, h_a commute with d_b, h_b, are exactly (11)–(12) for d_a^{-1}, h_a, e_i, f_i: for instance $d_a^{-1} e_i = q^{\eta_{ai}} e_i d_a^{-1}$ because d_a^{-1} commutes with h_{i+1}^{-1} and $d_a^{-1} y_i = q^{\eta_{ai}} y_i d_a^{-1}$ by (39). For (13) with $i \neq j$, rewrite $e_i f_j = -s^{-2} h_{i+1}^{-1} y_i x_j d_{j+1}^{-1}$ in the order $x_j d_{j+1}^{-1} h_{i+1}^{-1} y_i$ of the factors of $f_j e_i$; the three moves $y_i x_j \to x_j y_i$, $h_{i+1}^{-1} x_j \to x_j h_{i+1}^{-1}$, $y_i d_{j+1}^{-1} \to d_{j+1}^{-1} y_i$ contribute, by (40) and (39), the total exponent $(\delta_{j+1,i} - \delta_{j,i+1}) + \eta_{i+1,j} - \eta_{j+1,i} = 0$, so $e_i f_j = f_j e_i$. For $i = j$, the internal relation (24) ($h_{i+1}^{-1} y_i = q\, y_i h_{i+1}^{-1}$) and the mixed relations (39) ($d_{i+1}^{-1} y_i = q^{-1} y_i d_{i+1}^{-1}$, $h_{i+1}^{-1} x_i = q^{-1} x_i h_{i+1}^{-1}$) give, the powers of q cancelling, $e_i f_i = -s^{-2}\, y_i x_i\, h_{i+1}^{-1} d_{i+1}^{-1}$ and $f_i e_i = -s^{-2}\, x_i y_i\, h_{i+1}^{-1} d_{i+1}^{-1}$. Hence, by (41),

$$[e_i, f_j] = \delta_{ij}\, s^{-2} [x_i, y_i]\, h_{i+1}^{-1} d_{i+1}^{-1} = \delta_{ij}\, \frac{h_i h_{i+1}^{-1} - d_i d_{i+1}^{-1}}{s} = \Psi\Big(\delta_{ij} \frac{K_i^+ - (K_i^-)^{-1}}{s}\Big), \tag{42}$$

since $\Psi(K_i^+) = h_i h_{i+1}^{-1}$ and $\Psi((K_i^-)^{-1}) = d_{i+1}^{-1} d_i$. By Proposition 2.1, $\Psi : \mathcal{D} \to \overline{\mathcal{S}}_n(\mathfrak{B}^*)$ is a well-defined algebra homomorphism.

Step 4: bijectivity. On underlying vector spaces Ψ is the composite of three linear isomorphisms,

$$\mathcal{D} \xrightarrow{\sim} U_q(\mathfrak{b}_-) \otimes U_q(\mathfrak{b}_+) \xrightarrow[\text{Lemma 4.1}]{\sim} A_\ell \otimes A_u \xrightarrow[\text{Lemma 3.1}]{\sim} \overline{\mathcal{S}}_n(\mathfrak{B}^*), \tag{43}$$

the first being the inverse of multiplication in the double and the last the multiplication map (28): indeed, by Step 1, Ψ restricted to $U_q(\mathfrak{b}_-)$ is (33) followed by $f \mapsto f(B)$, and restricted to $U_q(\mathfrak{b}_+)$ it is (32) followed by $g \mapsto g(C)$, so $\Psi(wu) = \Psi(w)\Psi(u)$ with $\Psi(w)$ in the image of A_ℓ and $\Psi(u)$ in the image of A_u. Hence Ψ is bijective and $\Phi := \Psi^{-1}$ is an algebra isomorphism satisfying (16) and the case $j = i+1$ of (17).

Step 5: the formula for Φ on all arcs. It remains to check (17) for $j > i+1$; the entries b_{ij}, c_{ji} with $i < j$ vanish by (26). By (25) applied to C and B, and since h_k and d_k commute with c_{ij} and b_{ji} for $i < k < j$,

$$c_{ij} = s^{-1}h_k^{-1}[c_{ik}, c_{kj}], \qquad b_{ji} = s^{-1}d_k^{-1}[b_{ki}, b_{jk}] \qquad (i < k < j). \tag{44}$$

Take $k = j-1$ and argue by induction on $j - i$. In the first recurrence, substituting $\Phi(c_{ik}) = sH_k^+E_{ik}$ and $\Phi(c_{kj}) = sH_j^+E_k$ and moving the toral factors to the left (H_j^+ commutes with E_{ik}, and $E_kH_k^+ = q^{-1}H_k^+E_k$) gives $\Phi(c_{ij}) = s\,(H_k^+)^{-1}H_k^+H_j^+\big(E_{ik}E_k - q^{-1}E_kE_{ik}\big) = s\,H_j^+E_{ij}$. In the second, substituting $\Phi(b_{ki}) = -sF_{ki}(H_k^-)^{-1}$ and $\Phi(b_{jk}) = -sF_k(H_j^-)^{-1}$, moving $(H_k^-)^{-1}$ past F_k introduces a factor q by (5), $(H_j^-)^{-1}$ commutes with F_{ki}, and H_k^- commutes with $F_{ki}F_k$ and F_kF_{ki} (their weight is $\epsilon_j - \epsilon_i$, on which H_k^- acts trivially), so that

$$\Phi(b_{ji}) = s\big(qF_{ki}F_k - F_kF_{ki}\big)(H_j^-)^{-1} = -s\,F_{ji}(H_j^-)^{-1}$$

by (15). This proves (17) and completes part (1) of the theorem. □

6 The diagonal quotient

We now prove part (2). Under Φ the generators of I become

$$b_{aa}c_{aa} - 1 \longmapsto (H_a^-)^{-1}H_a^+ - 1. \tag{45}$$

The elements $z_a = (H_a^-)^{-1}H_a^+$ are central in $\mathcal{D}$: the two tori commute, and by (3), (5) and (12) the elements H_a^- and H_a^+ have the same conjugation action on every E_i and on every F_i. Since the $H_a^\pm$ are units, the ideal generated by the $z_a - 1$ is $\Phi(I) = J := (H_a^+ - H_a^- \mid 1 \le a \le n)$, so $\overline{\mathcal{S}}_n(\mathfrak{B}^*)/I \cong \mathcal{D}/J$. Recovering a quantum group from the double of its Borel algebras by identifying the two tori is the standard construction [5, Section 13], [7, Section 8.2.4]; we spell it out for the presentation in use. In $\mathcal{D}/J$ write H_a for the common image of $H_a^\pm$ and $K_i = H_iH_{i+1}^{-1}$. The presentation of Proposition 2.1 becomes: the relations (2) for the H_a, $H_aE_iH_a^{-1} = q^{\eta_{ai}}E_i$, $H_aF_iH_a^{-1} = q^{-\eta_{ai}}F_i$, the Serre relations, and $[E_i, F_j] = \delta_{ij}(K_i - K_i^{-1})/s$, while (11)–(12) become redundant. This is the presentation of the simply-connected $U_q(\mathfrak{sl}_n)$ given in Section 2, so $\mathcal{D}/J \cong U_q(\mathfrak{sl}_n)$ with $H_a^\pm \mapsto H_a$, $E_i \mapsto E_i$, $F_i \mapsto F_i$. Thus $\overline{\mathcal{S}}_n(\mathfrak{B}^*)/I \cong \mathcal{D}/J \cong U_q(\mathfrak{sl}_n)$; composing with Ψ and reading off (18)–(19) modulo I gives the explicit correspondences stated at the end of Theorem 2.2. That the sign and toral factors of (13) reproduce the standard relation (6) is a consistency check on the conventions of Section 2, not a derivation of them. □

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